

B2 Symmetry and Relativity — Model Solutions, 2020

Metric $(+, -, -, -)$ throughout. Capital bold $\mathbf{U}$, $\mathbf{P}$, etc. denote 4-vectors; lower-case bold $\mathbf{v}$, $\mathbf{p}$, etc. denote 3-vectors.

Question 1 — Threshold collisions and π^0 decay

Problem. A particle of mass m , total energy E , hits a stationary mass M , producing N particles of masses m_i . (a) Find threshold E . (b) Apply to $p + p \rightarrow p + p + \pi^0$: threshold proton kinetic energy and pion lab velocity. (c) Wavelength of the $\pi^0 \rightarrow 2\gamma$ photons in the pion rest frame; show half the photons go into a lab cone of half-angle $\cos^{-1}(v/c)$. (d) How can the threshold be lowered.

(a)

Total 4-momentum: $\mathbf{P}_{\text{tot}} = \mathbf{P}_m + \mathbf{P}_M$, conserved.

Lab frame, target at rest:

$$\mathbf{P}_M = (Mc, \mathbf{0}), \quad \mathbf{P}_m = (E/c, \mathbf{p}).$$

Sum the 4-momenta:

$$\mathbf{P}_{\text{tot}} = (E/c + Mc, \mathbf{p}).$$

Form the invariant square (metric $(+, -, -, -)$):

$$\mathbf{P}_{\text{tot}}^2 = (E/c + Mc)^2 - \mathbf{p}^2.$$

Expand the cross term $2(E/c)(Mc) = 2EM$:

$$\mathbf{P}_{\text{tot}}^2 = E^2/c^2 + 2EM + M^2c^2 - \mathbf{p}^2.$$

Apply on-shell projectile $E^2/c^2 - \mathbf{p}^2 = m^2c^2$:

$$\mathbf{P}_{\text{tot}}^2 = m^2c^2 + M^2c^2 + 2EM. \tag{1}$$

At threshold all products are at rest in the CoM frame:

$$\mathbf{P}_{\text{tot}}^2 = \left(\sum_i m_i\right)^2 c^2.$$

Equate to (1):

$$\left(\sum_i m_i\right)^2 c^2 = m^2c^2 + M^2c^2 + 2EM.$$

Isolate $2EM$:

$$2EM = \left[\left(\sum_i m_i\right)^2 - m^2 - M^2\right] c^2.$$

Divide by $2M$:

$$E_{\text{min}} = \frac{\left[\left(\sum_i m_i\right)^2 - m^2 - M^2\right] c^2}{2M}. \tag{2}$$

(b)

Set $m = M = m_p$, $\sum m_i = 2m_p + m_\pi$ in (2):

$$E_{\min} = \frac{[(2m_p + m_\pi)^2 - 2m_p^2]c^2}{2m_p}.$$

Expand $(2m_p + m_\pi)^2$:

$$(2m_p + m_\pi)^2 = 4m_p^2 + 4m_p m_\pi + m_\pi^2.$$

Substitute and collect:

$$E_{\min} = \frac{[2m_p^2 + 4m_p m_\pi + m_\pi^2]c^2}{2m_p}.$$

Divide term by term:

$$E_{\min} = \left(m_p + 2m_\pi + \frac{m_\pi^2}{2m_p}\right)c^2.$$

Form kinetic energy $T = E - m_p c^2$:

$$T_{\min} = \left(2m_\pi + \frac{m_\pi^2}{2m_p}\right)c^2.$$

Plug in $m_\pi c^2 = 135 \text{ MeV}$, $m_p c^2 = 938 \text{ MeV}$:

$$T_{\min} = 2(135) + \frac{135^2}{2(938)} \text{ MeV}.$$

Evaluate the second term:

$$\frac{135^2}{2(938)} = \frac{18225}{1876} \approx 9.7 \text{ MeV}.$$

Sum the contributions:

$$T_{\min} = 270 + 9.7 \text{ MeV}.$$

$T_{\min} \approx 280 \text{ MeV} \approx 2.07 m_\pi c^2.$

Comment: only $\sim 5\%$ above the naive $2m_\pi c^2$; the excess is the recoil kinetic energy of the products.

Pion lab velocity at threshold: all three products move together with the CoM velocity. Total 4-momentum:

$$\mathbf{P}_{\text{tot}} = (E/c + m_p c, \mathbf{p}).$$

Magnitude of 3-momentum from on-shell projectile:

$$|\mathbf{p}|c = \sqrt{E^2 - m_p^2 c^4}.$$

CoM velocity from $\mathbf{v} = c^2 \mathbf{p}_{\text{tot}} / E_{\text{tot}}$:

$$v_\pi = \frac{|\mathbf{p}|c^2}{E + m_p c^2}.$$

Total projectile energy $E = T_{\min} + m_p c^2$:

$$E = 280 + 938 = 1218 \text{ MeV}.$$

Compute $|\mathbf{p}|c$:

$$|\mathbf{p}|c = \sqrt{1218^2 - 938^2} \text{ MeV} = 777 \text{ MeV}.$$

Denominator:

$$E + m_p c^2 = 1218 + 938 = 2156 \text{ MeV}.$$

Divide:

$v_\pi/c = 777/2156 \approx 0.360.$

(c)

In the pion rest frame, $\pi^0 \rightarrow \gamma\gamma$ gives back-to-back photons. Energy conservation:

$$E'_\gamma = m_\pi c^2 / 2.$$

Photon wavelength from $E'_\gamma = hc/\lambda'$:

$$\lambda' = \frac{hc}{E'_\gamma} = \frac{2hc}{m_\pi c^2}.$$

Insert $hc = 1240 \text{ MeV fm}$:

$$\lambda' = \frac{2 \cdot 1240}{135} \text{ fm}.$$

Evaluate:

$$\boxed{\lambda' \approx 18.4 \text{ fm.}}$$

Cone of half the photons. Isotropy in pion frame: half are emitted into the forward hemisphere $\theta' \in [0, \pi/2]$.

Aberration formula (boost along $+x$, θ' measured from $+x$ in pion frame):

$$\cos \theta = \frac{\cos \theta' + \beta}{1 + \beta \cos \theta'}, \quad \beta = v/c.$$

Set the hemisphere boundary $\theta' = \pi/2$:

$$\cos \theta' = 0.$$

Substitute:

$$\cos \theta = \frac{0 + \beta}{1 + 0} = \beta.$$

Invert:

$$\boxed{\theta = \cos^{-1}(v/c)}.$$

Half the photons (those with $\cos \theta' \geq 0$) lie inside this lab cone.

(d)

Use a colliding-beam (symmetric) configuration: two protons of equal and opposite 3-momentum, so the lab *is* the CoM frame.

Threshold in CoM frame: total 4-momentum has zero 3-momentum, so

$$\mathbf{P}_{\text{tot}}^2 = (2E_{\text{lab}}/c)^2.$$

Set equal to the threshold invariant $(2m_p + m_\pi)^2 c^2$:

$$(2E_{\text{lab}}/c)^2 = (2m_p + m_\pi)^2 c^2.$$

Solve for E_{lab} :

$$E_{\text{lab}} = (m_p + m_\pi/2)c^2.$$

Kinetic energy per beam:

$$T_{\text{beam}} = E_{\text{lab}} - m_p c^2 = \frac{1}{2} m_\pi c^2 = 67.5 \text{ MeV}.$$

Total kinetic energy (both beams):

$$T_{\text{total}} = 2T_{\text{beam}} = m_\pi c^2 = 135 \text{ MeV}.$$

Compare to fixed-target value:

$$\boxed{\Delta T = 280 - 135 = 145 \text{ MeV saved; reduction factor } \approx 2.}$$

Final state: at threshold the three products p, p, π^0 are at rest in the lab.

Question 2 — Rocket dynamics and the photon rocket

Problem. (a) Define $\mathbf{U}$, $\mathbf{F}$ and show $dE/dt = \mathbf{v} \cdot \mathbf{f}$. (b) Rocket with proper acceleration a_0 : write equation of motion in rapidity η , derive trajectory $x^2 - c^2 t^2 = (c^2/a_0)^2$, and show a photon emitted from origin never catches the rocket. (c) Photon rocket: show $m \rightarrow \alpha m$ with $\alpha = \sqrt{(1-\beta)/(1+\beta)}$; fuel fraction to reach $\gamma = 8$ then decelerate; comment on massive-exhaust case.

(a)

Definitions:

$$\mathbf{U} = \frac{d\mathbf{X}}{d\tau} = \gamma(c, \mathbf{v}), \quad \mathbf{F} = \frac{d\mathbf{P}}{d\tau} = \gamma(\dot{E}/c, \mathbf{f}),$$

where $\mathbf{P} = (E/c, \mathbf{p})$, $\mathbf{f} = d\mathbf{p}/dt$ is the 3-force, τ proper time.

Pure force preserves rest mass: $\mathbf{P}^2 = m^2 c^2 = \text{const.}$ Differentiate w.r.t. proper time τ :

$$2\mathbf{P} \cdot \mathbf{F} = 0 \Rightarrow \mathbf{P} \cdot \mathbf{F} = 0.$$

Contract with the metric ($\mathbf{P} \cdot \mathbf{F} = P^0 F^0 - \mathbf{p} \cdot (\gamma \mathbf{f})$):

$$\frac{E}{c} \cdot \gamma \frac{\dot{E}}{c} - \mathbf{p} \cdot \gamma \mathbf{f} = 0.$$

Cancel the overall γ and write $\dot{E} \equiv dE/dt$:

$$\frac{E}{c^2} \frac{dE}{dt} - \mathbf{p} \cdot \mathbf{f} = 0.$$

Use $\mathbf{p} = (E/c^2)\mathbf{v}$:

$$\frac{E}{c^2} \frac{dE}{dt} = \frac{E}{c^2} \mathbf{v} \cdot \mathbf{f}.$$

Cancel E/c^2 :

$$\frac{dE}{dt} = \mathbf{v} \cdot \mathbf{f}.$$

(b)

Rapidity definition: $\cosh \eta = \gamma$, $\sinh \eta = \gamma\beta$, so $v/c = \tanh \eta$.

In the instantaneous rest frame the 4-acceleration is purely spatial with magnitude a_0 :

$$|A^\mu A_\mu| = a_0^2.$$

(Strictly spacelike with $(+, -, -, -)$, so $A^\mu A_\mu = -a_0^2$; we equate magnitudes.) Use the stated hint:

$$c^2 \dot{\eta}^2 \cosh^2 \eta = a_0^2.$$

Take the positive root:

$$c \dot{\eta} \cosh \eta = a_0.$$

Recognise $\dot{\eta} \cosh \eta = d(\sinh \eta)/dt$:

$$c \frac{d}{dt} \sinh \eta = a_0.$$

Integrate with $\eta(0) = 0$:

$$c \sinh \eta = a_0 t.$$

$$\sinh \eta = a_0 t / c. \quad (3)$$

Lab velocity from $v = c \tanh \eta = c \sinh \eta / \cosh \eta$:

$$v = \frac{a_0 t}{\sqrt{1 + (a_0 t / c)^2}}.$$

Position from $dx/dt = v$:

$$x(t) = \int_0^t \frac{a_0 t' dt'}{\sqrt{1 + (a_0 t' / c)^2}}.$$

Substitute $u = 1 + (a_0 t' / c)^2$, $du = 2(a_0^2 / c^2) t' dt'$:

$$x(t) = \frac{c^2}{2a_0} \int_1^{1+(a_0 t/c)^2} u^{-1/2} du.$$

Evaluate:

$$x(t) = \frac{c^2}{a_0} [\sqrt{1 + (a_0 t / c)^2} - 1].$$

Choose origin at $x_0 = -c^2/a_0$ (shift $x \rightarrow x + c^2/a_0$):

$$x(t) = \frac{c^2}{a_0} \sqrt{1 + (a_0 t / c)^2}.$$

Square both sides:

$$x^2 = (c^2/a_0)^2 + c^2 t^2.$$

Rearrange:

$$\boxed{x^2 - c^2 t^2 = (c^2/a_0)^2}. \quad (4)$$

Photon never catches the rocket. Photon worldline from $(t, x) = (0, 0)$ in $+x$:

$$x_\gamma = ct.$$

Rocket worldline from (4):

$$x_{\text{rocket}}(t) = \sqrt{(c^2/a_0)^2 + c^2 t^2}.$$

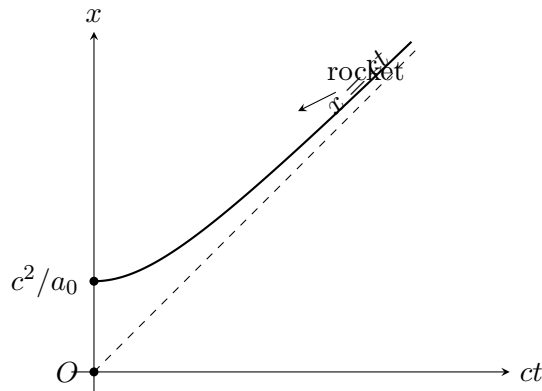
Compare termwise:

$$x_{\text{rocket}}^2 - x_\gamma^2 = (c^2/a_0)^2 > 0 \quad \forall t,$$

so $x_{\text{rocket}} > x_\gamma$ for all $t \geq 0$. Asymptotic gap at large t via $\sqrt{1 + \epsilon} \approx 1 + \epsilon/2$:

$$x_{\text{rocket}} - ct = ct \left[\sqrt{1 + (c/a_0 t)^2} - 1 \right] \approx \frac{c^3}{2a_0^2 t} \rightarrow 0^+.$$

The line $x = ct$ is the asymptote of the hyperbola: a *Rindler horizon* bounding the rocket's past light cone of accessible signals from the origin.



(c)

Initial state (lab frame): rocket of mass m at rest, $\mathbf{P}_i = (mc, \mathbf{0})$.

After emission: rocket has mass m' , lab speed v along $+\hat{\mathbf{v}}$, 4-momentum $\mathbf{P}_r = \gamma m'(c, \mathbf{v})$. Photons travel backward:

$$\mathbf{P}_\gamma = (E_\gamma/c, -(E_\gamma/c)\hat{\mathbf{v}}).$$

Energy conservation:

$$mc = \gamma m'c + E_\gamma/c.$$

Momentum conservation along $\hat{\mathbf{v}}$:

$$0 = \gamma m'v - E_\gamma/c.$$

Solve the second for E_γ/c :

$$E_\gamma/c = \gamma m'v.$$

Substitute into the first:

$$mc = \gamma m'c + \gamma m'v.$$

Factor $\gamma m'$ and divide by c :

$$m = \gamma m'(1 + \beta).$$

Solve for m'/m :

$$\frac{m'}{m} = \frac{1}{\gamma(1 + \beta)}.$$

Use $1/\gamma = \sqrt{1 - \beta^2} = \sqrt{(1 - \beta)(1 + \beta)}$:

$$\frac{m'}{m} = \frac{\sqrt{(1 - \beta)(1 + \beta)}}{1 + \beta}.$$

Simplify:

$$\boxed{\alpha \equiv \frac{m'}{m} = \sqrt{\frac{1 - \beta}{1 + \beta}}.} \quad (5)$$

Fuel fraction. Two stages of equal $|\beta_1|$: accelerate to β_1 (factor α_1), then decelerate from β_1 to rest (factor α_1 again, applied in the instantaneous rest frame of the rocket at the start of deceleration — the derivation depends only on $|\beta|$, not its sign):

$$m_{\text{final}}/m_{\text{initial}} = \alpha_1 \cdot \alpha_1 = \alpha_1^2.$$

Compute β_1 from $\gamma_1 = 8$:

$$\beta_1 = \sqrt{1 - 1/\gamma_1^2} = \sqrt{1 - 1/64}.$$

Evaluate:

$$\beta_1 = \sqrt{63}/8 \approx 0.9922.$$

Form numerator and denominator:

$$1 - \beta_1 \approx 0.00784, \quad 1 + \beta_1 \approx 1.9922.$$

Divide:

$$\alpha_1^2 = \frac{1 - \beta_1}{1 + \beta_1} \approx \frac{0.00784}{1.9922} \approx 3.93 \times 10^{-3}.$$

Fuel fraction:

$$\boxed{f = 1 - \alpha_1^2 \approx 0.996, \text{ i.e. } \approx 99.6\% \text{ of initial mass is fuel.}}$$

Massive exhaust. Photons carry maximal momentum per unit energy ($p = E/c$). Massive exhaust at the same energy carries $p < E/c$, so a larger expended rest-mass is needed for the same impulse: the fuel fraction increases.

Question 3 — Field tensor, gauge invariance, moving wire

Problem. (a) Components of $F^{\mu\nu}$; show $\partial_\nu F^{\mu\nu} = \mu_0 J^\mu$ encodes Gauss + Ampère; build an invariant of $\mathbf{E}, \mathbf{B}$. (b) From $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$ show gauge invariance and reduce Maxwell to $\partial_\nu \partial^\nu A^\mu = -\mu_0 J^\mu$ in a suitable gauge. (c) Charged wire moving along its length: find $\mathbf{E}, \mathbf{B}$ in rest frame S and lab S' ; is there a frame with $\mathbf{E} = 0$?

(a)

Field tensor (metric $(+, -, -, -)$, raised indices; standard convention):

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix},$$

i.e. $F^{0i} = -E_i/c$, $F^{ij} = -\epsilon^{ijk} B_k$. 4-current: $J^\mu = (c\rho, \mathbf{J})$. (The question's equation $\partial_\nu F^{\mu\nu} = \mu_0 J^\mu$ contracts on the second index; by antisymmetry $\partial_\nu F^{\mu\nu} = -\partial_\nu F^{\nu\mu}$, equivalent up to a sign to the canonical Steane form $\partial_\mu F^{\mu\nu} = \mu_0 J^\nu$. Both yield Maxwell — we work with the canonical form below.)

Take $\nu = 0$ in $\partial_\mu F^{\mu\nu} = \mu_0 J^\nu$:

$$\partial_\mu F^{\mu 0} = \mu_0 J^0.$$

Antisymmetry kills $\mu = 0$:

$$\partial_i F^{i0} = \mu_0 c\rho.$$

Insert $F^{i0} = E_i/c$:

$$\frac{1}{c} \nabla \cdot \mathbf{E} = \mu_0 c\rho.$$

Use $\mu_0 c^2 = 1/\epsilon_0$:

$$\nabla \cdot \mathbf{E} = \rho/\epsilon_0. \quad \checkmark (\text{Gauss})$$

Take $\nu = i$ (spatial):

$$\partial_\mu F^{\mu i} = \partial_0 F^{0i} + \partial_j F^{ji} = \mu_0 J^i.$$

Insert $F^{0i} = -E_i/c$ and $F^{ji} = -F^{ij} = +\epsilon^{ijk} B_k$:

$$-\frac{1}{c^2} \partial_t E_i + \epsilon^{ijk} \partial_j B_k = \mu_0 J^i.$$

Identify $\epsilon^{ijk} \partial_j B_k = (\nabla \times \mathbf{B})_i$:

$$(\nabla \times \mathbf{B})_i - \frac{1}{c^2} \partial_t E_i = \mu_0 J^i.$$

Promote to vector form:

$$\nabla \times \mathbf{B} - \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \mathbf{J}. \quad \checkmark (\text{Ampère–Maxwell})$$

Invariant. Form the scalar $F_{\mu\nu} F^{\mu\nu}$ by lowering indices with $\eta = \text{diag}(+, -, -, -)$:

$$F_{\mu\nu} F^{\mu\nu} = 2 \sum_i F^{0i} F_{0i} + \sum_{i,j} F^{ij} F_{ij}.$$

Compute the time-space piece (one index lowered flips sign): with $F^{0i} = E_i/c$, $F_{0i} = -E_i/c$,

$$\sum_i F^{0i} F_{0i} = -E^2/c^2.$$

Compute the space-space piece using $F^{ij} = -\epsilon^{ijk} B_k$ and $F_{ij} = F^{ij}$:

$$\sum_{i,j} F^{ij} F_{ij} = 2B^2.$$

Sum the contributions:

$$F_{\mu\nu} F^{\mu\nu} = -2E^2/c^2 + 2B^2.$$

Halve to isolate the invariant:

$$\boxed{\frac{1}{2} F_{\mu\nu} F^{\mu\nu} = B^2 - E^2/c^2 = \text{Lorentz invariant.}}$$

(b)

Gauge transformation $A^\mu \rightarrow A^\mu + \partial^\mu \chi$ for any scalar χ :

$$F^{\mu\nu} \rightarrow \partial^\mu (A^\nu + \partial^\nu \chi) - \partial^\nu (A^\mu + \partial^\mu \chi).$$

Distribute the derivatives:

$$F^{\mu\nu} \rightarrow F^{\mu\nu} + \partial^\mu \partial^\nu \chi - \partial^\nu \partial^\mu \chi.$$

The mixed partials commute:

$$\partial^\mu \partial^\nu \chi - \partial^\nu \partial^\mu \chi = 0.$$

Therefore:

$$F^{\mu\nu} \rightarrow F^{\mu\nu}. \quad \checkmark$$

Equivalent 3-vector check: with $\phi \rightarrow \phi + \partial_t \chi$ and $\mathbf{A} \rightarrow \mathbf{A} - \nabla \chi$,

$$\mathbf{B} = \nabla \times \mathbf{A} \rightarrow \nabla \times \mathbf{A} - \nabla \times \nabla \chi = \mathbf{B}.$$

$$\mathbf{E} = -\nabla \phi - \partial_t \mathbf{A} \rightarrow -\nabla \phi - \partial_t \nabla \chi - \partial_t \mathbf{A} + \partial_t \nabla \chi = \mathbf{E}.$$

Substitute $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$ into $\partial_\nu F^{\mu\nu} = \mu_0 J^\mu$:

$$\partial_\nu \partial^\mu A^\nu - \partial_\nu \partial^\nu A^\mu = \mu_0 J^\mu.$$

Swap derivative order in the first term (partials commute):

$$\partial^\mu (\partial_\nu A^\nu) - \partial_\nu \partial^\nu A^\mu = \mu_0 J^\mu.$$

Impose Lorenz gauge:

$$\partial_\nu A^\nu = 0.$$

The first term vanishes, leaving:

$$\boxed{\partial_\nu \partial^\nu A^\mu = -\mu_0 J^\mu.}$$

Note: $\partial_\nu \partial^\nu = (1/c^2) \partial_t^2 - \nabla^2 = \square$, so this is $\square A^\mu = -\mu_0 J^\mu$; the more common form $\square A^\mu = \mu_0 J^\mu$ corresponds to the alternate sign convention on $F^{\mu\nu}$ noted in (a).

(c)

Wire rest frame S : linear charge density ρ , no current. Apply Gauss to a cylinder of radius r and length L :

$$2\pi r L E_r = \rho L / \epsilon_0.$$

Solve for E_r :

$$E_r = \frac{\rho}{2\pi\epsilon_0 r}.$$

$$\boxed{\mathbf{E}_S = \frac{\rho}{2\pi\epsilon_0 r} \hat{\mathbf{r}}, \quad \mathbf{B}_S = \mathbf{0}.$$

Lab frame S' : the wire moves at $\mathbf{v} = v\hat{\mathbf{z}}$ in S' , so S' moves at $-\mathbf{v} = -v\hat{\mathbf{z}}$ relative to S . Field transformation for S' moving with velocity $\mathbf{u}$ relative to S ($\mathbf{u} = -\mathbf{v}$):

$$\mathbf{E}'_{\perp} = \gamma(\mathbf{E}_{\perp} + \mathbf{u} \times \mathbf{B}_{\perp}), \quad \mathbf{B}'_{\perp} = \gamma(\mathbf{B}_{\perp} - (\mathbf{u}/c^2) \times \mathbf{E}_{\perp}).$$

Insert $\mathbf{B} = \mathbf{0}$ and $\mathbf{u} = -\mathbf{v}$:

$$\mathbf{E}' = \gamma \mathbf{E} = \frac{\gamma \rho}{2\pi\epsilon_0 r} \hat{\mathbf{r}}.$$

$$\mathbf{B}' = -\gamma(\mathbf{u}/c^2) \times \mathbf{E} = +\gamma(\mathbf{v}/c^2) \times E_r \hat{\mathbf{r}}.$$

Use $\mathbf{v} = v\hat{\mathbf{z}}$ and $\hat{\mathbf{z}} \times \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}}$:

$$\mathbf{B}' = \frac{\gamma \rho v}{2\pi\epsilon_0 c^2 r} \hat{\boldsymbol{\phi}}.$$

Apply $1/(\epsilon_0 c^2) = \mu_0$:

$$\mathbf{B}' = \frac{\mu_0 \gamma \rho v}{2\pi r} \hat{\boldsymbol{\phi}}.$$

Identify lab-frame line density $\rho' = \gamma \rho$ (length contraction) and current $I' = \rho' v = \gamma \rho v$:

$$\mathbf{E}' = \frac{\rho'}{2\pi\epsilon_0 r} \hat{\mathbf{r}}, \quad \mathbf{B}' = \frac{\mu_0 I'}{2\pi r} \hat{\boldsymbol{\phi}}.$$

$$\boxed{\mathbf{E}' \text{ Coulomb of } \rho' = \gamma \rho; \quad \mathbf{B}' \text{ Ampère of } I' = \gamma \rho v.}$$

Frame with $\mathbf{E} = 0$? Use the invariant $I_1 \equiv B^2 - E^2/c^2$. Evaluate in S :

$$I_1|_S = 0 - E_S^2/c^2 = -E_S^2/c^2 < 0.$$

Lorentz invariance forces $I_1 < 0$ in every frame:

$$B^2 - E^2/c^2 < 0 \Rightarrow E^2 > c^2 B^2 \geq 0 \Rightarrow \mathbf{E} \neq \mathbf{0}.$$

$$\boxed{\text{No frame has } \mathbf{E} = \mathbf{0}.$$

Question 4 — EM wave 4-potential, intervals, photon CoM, relative speed

Problem. (a) Write A^μ for a vacuum EM wave; show phase ϕ is invariant; show an isolated electron cannot emit a photon. (b) Condition for spacelike separation; can simultaneity be achieved? (c) Two photons with 4-momenta $\mathbf{P}_1$ ($+x$), $\mathbf{P}_2$ ($+y$), same ω : rest mass and CoM velocity. (d) Derive the relative speed formula.

(a)

4-potential of a vacuum plane wave:

$$A^\mu(\mathbf{X}) = A_0^\mu \cos(\mathbf{K} \cdot \mathbf{X}),$$

with constant polarisation A_0^μ , 4-wavevector $\mathbf{K} = (\omega/c, \mathbf{k})$ (null: $\mathbf{K}^2 = 0$ in vacuum) and event $\mathbf{X} = (ct, \mathbf{x})$.

Write the phase using the metric:

$$\phi = \mathbf{K} \cdot \mathbf{X} = K_\mu X^\mu = \omega t - \mathbf{k} \cdot \mathbf{x}.$$

$K_\mu X^\mu$ is a contraction of two 4-vectors:

$\phi = K_\mu X^\mu$ is Lorentz invariant.

Electron $\rightarrow e + \gamma$ **forbidden.** 4-momentum conservation:

$$\mathbf{P}_e = \mathbf{P}'_e + \mathbf{P}_\gamma.$$

Isolate the photon and square:

$$\mathbf{P}_\gamma = \mathbf{P}_e - \mathbf{P}'_e.$$

Take the invariant square:

$$\mathbf{P}_\gamma^2 = \mathbf{P}_e^2 - 2\mathbf{P}_e \cdot \mathbf{P}'_e + \mathbf{P}'_e{}^2.$$

Insert $\mathbf{P}_e^2 = \mathbf{P}'_e{}^2 = m_e^2 c^2$ and $\mathbf{P}_\gamma^2 = 0$:

$$0 = 2m_e^2 c^2 - 2\mathbf{P}_e \cdot \mathbf{P}'_e.$$

Solve for the dot product:

$$\mathbf{P}_e \cdot \mathbf{P}'_e = m_e^2 c^2.$$

Evaluate in the initial electron rest frame, $\mathbf{P}_e = (m_e c, \mathbf{0})$, $\mathbf{P}'_e = (E'_e/c, \mathbf{p}'_e)$:

$$\mathbf{P}_e \cdot \mathbf{P}'_e = m_e c \cdot E'_e/c = m_e E'_e.$$

Equate:

$$m_e E'_e = m_e^2 c^2.$$

Cancel one m_e :

$$E'_e = m_e c^2.$$

But $E'_e = \sqrt{m_e^2 c^4 + |\mathbf{p}'_e|^2 c^2} \geq m_e c^2$, with equality only at $\mathbf{p}'_e = \mathbf{0}$. Then $\mathbf{P}_\gamma = \mathbf{P}_e - \mathbf{P}'_e = 0$, i.e. no photon:

No photon can be emitted; $e^- \rightarrow e^- \gamma$ forbidden in vacuum.

(b)

Squared interval with $(+, -, -, -)$:

$$(\Delta \mathbf{X})^2 = c^2(\Delta t)^2 - (\Delta \mathbf{x})^2.$$

Spacelike means $(\Delta \mathbf{X})^2 < 0$:

$$c^2(t_2 - t_1)^2 < (\mathbf{x}_2 - \mathbf{x}_1)^2.$$

Simultaneity in S' : require $\Delta t' = 0$. Lorentz boost along $\hat{\mathbf{n}}$ with speed u :

$$c\Delta t' = \gamma(c\Delta t - (u/c) \hat{\mathbf{n}} \cdot \Delta \mathbf{x}).$$

Set $\Delta t' = 0$:

$$c\Delta t = (u/c) \hat{\mathbf{n}} \cdot \Delta \mathbf{x}.$$

Choose $\hat{\mathbf{n}}$ along $\Delta \mathbf{x}$:

$$c\Delta t = (u/c) |\Delta \mathbf{x}|.$$

Solve for u :

$$u = \frac{c^2 \Delta t}{|\Delta \mathbf{x}|}.$$

Spacelike $\Rightarrow c|\Delta t| < |\Delta \mathbf{x}| \Rightarrow |u| < c$, achievable:

Yes, a frame S' exists in which the events are simultaneous.

(c)

Photon 4-momenta (using $E = \hbar\omega$, $|\mathbf{p}|c = E$):

$$\mathbf{P}_1 = (\hbar\omega/c)(1, 1, 0, 0), \quad \mathbf{P}_2 = (\hbar\omega/c)(1, 0, 1, 0).$$

Sum component-wise:

$$\mathbf{P} = \mathbf{P}_1 + \mathbf{P}_2 = (\hbar\omega/c)(2, 1, 1, 0).$$

Square with $(+, -, -, -)$:

$$\mathbf{P}^2 = (\hbar\omega/c)^2(2^2 - 1^2 - 1^2 - 0^2).$$

Simplify:

$$\mathbf{P}^2 = 2(\hbar\omega/c)^2.$$

Identify rest mass via $\mathbf{P}^2 = M^2 c^2$:

$$M^2 c^2 = 2(\hbar\omega/c)^2.$$

Solve:

$$M = \sqrt{2} \hbar\omega/c^2.$$

CoM velocity from $\mathbf{v}_{\text{CoM}} = c^2 \mathbf{p}_{\text{tot}}/E_{\text{tot}}$:

$$\mathbf{p}_{\text{tot}} = (\hbar\omega/c)(\hat{\mathbf{x}} + \hat{\mathbf{y}}), \quad E_{\text{tot}} = 2\hbar\omega.$$

Substitute:

$$\mathbf{v}_{\text{CoM}} = \frac{c^2(\hbar\omega/c)(\hat{\mathbf{x}} + \hat{\mathbf{y}})}{2\hbar\omega}.$$

Cancel $\hbar\omega$:

$$\mathbf{v}_{\text{CoM}} = \frac{c}{2}(\hat{\mathbf{x}} + \hat{\mathbf{y}}).$$

Magnitude:

$$\mathbf{v}_{\text{CoM}} = \frac{c}{2}(\hat{\mathbf{x}} + \hat{\mathbf{y}}), \quad |\mathbf{v}_{\text{CoM}}| = c/\sqrt{2}.$$

(d)

Particle 1 has 3-velocity $\mathbf{u}$ in S ; particle 2 has velocity $\mathbf{v}$ in S . Boost from S to particle 2's rest frame S' (which moves at $\mathbf{v}$ in S); the relative speed is $w = |\mathbf{u}'|$.

Decompose $\mathbf{u}$ into components parallel and perpendicular to $\mathbf{v}$. Velocity transformation:

$$\mathbf{u}'_{\parallel} = \frac{\mathbf{u}_{\parallel} - \mathbf{v}}{1 - \mathbf{u} \cdot \mathbf{v}/c^2}, \quad \mathbf{u}'_{\perp} = \frac{\mathbf{u}_{\perp}}{\gamma_v(1 - \mathbf{u} \cdot \mathbf{v}/c^2)}.$$

Abbreviate $D = 1 - \mathbf{u} \cdot \mathbf{v}/c^2$.

Square magnitude (parallel and perpendicular pieces independent):

$$|\mathbf{u}'|^2 = |\mathbf{u}'_{\parallel}|^2 + |\mathbf{u}'_{\perp}|^2.$$

Insert the transformations:

$$|\mathbf{u}'|^2 = \frac{(\mathbf{u}_{\parallel} - \mathbf{v})^2}{D^2} + \frac{\mathbf{u}_{\perp}^2}{\gamma_v^2 D^2}.$$

Factor out $1/D^2$:

$$|\mathbf{u}'|^2 = \frac{1}{D^2} \left[(\mathbf{u}_{\parallel} - \mathbf{v})^2 + \mathbf{u}_{\perp}^2 / \gamma_v^2 \right].$$

Use $1/\gamma_v^2 = 1 - v^2/c^2$:

$$|\mathbf{u}'|^2 = \frac{(\mathbf{u}_{\parallel} - \mathbf{v})^2 + (1 - v^2/c^2)\mathbf{u}_{\perp}^2}{D^2}.$$

Expand the perpendicular piece:

$$(\mathbf{u}_{\parallel} - \mathbf{v})^2 + (1 - v^2/c^2)\mathbf{u}_{\perp}^2 = (\mathbf{u}_{\parallel} - \mathbf{v})^2 + \mathbf{u}_{\perp}^2 - (v^2/c^2)\mathbf{u}_{\perp}^2.$$

Combine $(\mathbf{u}_{\parallel} - \mathbf{v})^2 + \mathbf{u}_{\perp}^2 = (\mathbf{u} - \mathbf{v})^2$ (since $\mathbf{v}$ is along $\parallel$):

$$(\mathbf{u}_{\parallel} - \mathbf{v})^2 + (1 - v^2/c^2)\mathbf{u}_{\perp}^2 = (\mathbf{u} - \mathbf{v})^2 - (v^2/c^2)\mathbf{u}_{\perp}^2.$$

Apply the hint $(\mathbf{a} \times \mathbf{b})^2 = a^2 b^2 - (\mathbf{a} \cdot \mathbf{b})^2$ with $\mathbf{a} = \mathbf{v}$, $\mathbf{b} = \mathbf{u}$:

$$(\mathbf{v} \times \mathbf{u})^2 = v^2 u^2 - (\mathbf{u} \cdot \mathbf{v})^2.$$

Note $\mathbf{u} \cdot \mathbf{v} = u_{\parallel} v$ and $\mathbf{u}_{\perp}^2 = u^2 - u_{\parallel}^2$:

$$v^2 \mathbf{u}_{\perp}^2 = v^2 u^2 - v^2 u_{\parallel}^2 = v^2 u^2 - (\mathbf{u} \cdot \mathbf{v})^2 = (\mathbf{u} \times \mathbf{v})^2.$$

Substitute back:

$$|\mathbf{u}'|^2 = \frac{(\mathbf{u} - \mathbf{v})^2 - (\mathbf{u} \times \mathbf{v})^2/c^2}{D^2}.$$

Take the positive square root:

$$\boxed{w = |\mathbf{u}'| = \frac{\sqrt{(\mathbf{u} - \mathbf{v})^2 - (\mathbf{u} \times \mathbf{v})^2/c^2}}{1 - \mathbf{u} \cdot \mathbf{v}/c^2}}.$$

Sanity: parallel motion $\mathbf{u} \times \mathbf{v} = 0$ recovers $w = |u - v|/(1 - uv/c^2)$; non-relativistic limit $w \rightarrow |\mathbf{u} - \mathbf{v}|$.